Solving school problems in Python: review



Optimization · Level 3 · Unit check

In this unit

- A timetable that works
- Exams without clashes
- The school bus route
- Matching teachers and classes

The big ideas

- Say the rules as lines of a model; the solver crosses out and searches
- INFEASIBLE proves the rules clash; OPTIMAL proves the best for the goal you wrote
- Quick rules build plans fast; improvements and solvers make them better
- Judge every plan twice, total and worst case, with the people it affects

Model it in teams

1. Warm-up (5 min): Name one school decision for each lesson.
2. Model (15 min): Teams write options, rules and a goal.
3. Swap (10 min): Another team finds a missing rule.
4. Wrap-up (5 min): Who would each plan be hardest on?

No devices needed. If you have a phone, run one team's model in the "Model it with CP-SAT" lab at the end.

Let’s discuss

1. Which school decision would you model first, and why?
2. When is a quick rule good enough?
3. Who should see a model’s rules before it decides?

Exit ticket

- One school decision I will model, its goal, and one fairness rule.